Maize research at IAR Samaru

S.G. Ado[1], F.A. Showemimo[1], S.O. Alabi[1], B. Badu-Apraku[2],
A. Menkir[2], I.S. Usman[1] and U.S. Abdullahi[1]
[1]Institute for Agricultural Research, Samaru
Ahmadu Bello University, Zaria, Nigeria
[2]International Institute of Tropical Agriculture, Ibadan, Nigeria

Abstract
Maize (*Zea mays* L.) research at the Institute for Agricultural Research (IAR), Samaru was initiated in the 1950s with the goal of developing adapted high yielding varieties and hybrids. Germplasm materials were collected from local sources as well as from different countries, including Columbia, Kenya, Mexico and the US. By the 1960s, IAR Samaru was actively participating in the National Cooperative Trials, the West African Maize Variety Trials of the OAU-STRC Joint Project as well as the East African Maize Variety Trial coordinated by the East African Agricultural and Forestry Research Organization (EAAFRO), Kitale, Kenya. By 1976, a total of nine new open- pollinated varieties had been developed and released to farmers. The new maize technology packages developed gave up to 98% higher grain yield than the farmers' traditional methods of production. In the 1990s, collaborative research continued under the West and Central African Collaborative Maize Research Network (WECAMAN) and the Sasakawa Global 2000 (SG-2000) which, in partnership with IAR, disseminated new maize technologies to Nigerian farmers. In 2001, three new varieties developed and tested in collaboration with IITA-WECAMAN were registered and released to Nigerian farmers. Recently, Quality Protein Maize (QPM) trials were initiated in collaboration with IITA-WECAMAN, SG-2000, and CIMMYT and promising results have been obtained. The Institute continues to collaborate with national and international research organizations in maize improvement. The five decades of collaborative maize research at Samaru have contributed to increased land area under maize, increased grain yield, and increased income to maize farmers.

Résumé
La recherche sur le maïs à Samaru a été initiée dans les années 50 avec le projet d'amélioration variétale centré sur le développement d'hybrides et de variétés adaptés à hauts rendements. Les ressources génétiques comprenaient du matériel de source locale ainsi que des introductions de différents pays incluant la Columbie, le Kenya, le

Mexique et les USA. Dans les années 60, l'IAR à Samaru a participé aux essais nationaux de collaboration, au projet conjoint OAU-STRC des essais variétaux en Afrique de l'ouest ainsi qu'aux essais variétaux de maïs pour l'Afrique de l'est, coordonnés par l'Organisation de recherche en Agriculture et en foresterie de l'Afrique de l'est basée à Kitaly, Kenya. En 1976, un total de neuf nouvelles variétés à pollinisation libre a été développé et vulgarisé auprès des agriculteurs. Pendant les années 80, les variétés hybrides développées par l'IITA en collaboration avec des chercheurs nationaux ont été homologuées et commercialisées par des entreprises privées de semences. Les nouveaux paquets technologiques développés donnaient jusqu'à 98 pour cent en rendement grain plus élevé que les méthodes traditionnelles de production des agriculteurs. Dans les années 90, l'IAR à Samaru en partenariat avec le Réseau de recherche sur le maïs pour l'Afrique de l'ouest et du centre (WECAMAN) et avec Sasakawa Global 2000, ont continué la recherche collaborative avec la promotion de nouvelles technologies maïsicoles auprès des agriculteurs Nigérians. En 2001, trois nouvelles variétés développées et testées en collaboration avec l'IITA-WECAMAN ont été homologuées et vulgarisées auprès des agriculteurs Nigérians. Récemment, des essais de maïs riches en protéines de qualité ont été initiés en collaboration avec l'IITA-WECAMAN, SG2000 et le CIMMYT et des résultats prometteurs ont été obtenus. L'institut continue à coopérer avec les organisations de recherche internationale et nationale pour l'amélioration variétale du maïs. L'impact des cinq décennies de recherche collaborative à Samaru a induit l'accroissement des superficies emblavées en maïs, l'augmentation des rendements grains et l'augmentation des revenus des agriculteurs qui cultivent le maïs.

Background

The Institute for Agricultural Research, Samaru, was established in 1922 as the administrative headquarters of the Department of Agriculture of the defunct Northern Region of Nigeria. In 1962 it was affiliated to the Ahmadu Bello University and mandated to conduct research for all the important crops grown in the entire Northern Region, including maize (*Zea mays* L.). Research programs of the Institute developed technological packages for improved crop production. Each research program is composed of various research projects and subprojects. Maize research is under the Cereals Research Program and it focuses on varietal improvement, cultural practices and management, crop protection, socioeconomics of production and technology adoption.

Varietal Improvement

The Maize Varietal Improvement Project (MVIP) was initiated in the 1950s with the aim of developing adapted but genetically diverse populations. The maize idiotype for the ecologies of northern Nigeria was defined as follows: plants with good root development to prevent root lodging, tasseling in 55–65 days, ear height of 1.5 m or less, tight husk cover to prevent damage from insects and birds, and ear shanks that allow the ears to droop upon maturing so that water from the heavy rains in September would not penetrate into the ears through the tip of the husks (Anon. 1969).

In practical terms, the main aims of the MVIP were to develop high yielding varieties or hybrids that could be planted early in June, reach anthesis in early August, and mature shortly before the dry season sets in early in October. To achieve these objectives, germplasm materials were collected locally as well as being introduced from different sources. Several introductions from Columbia in the early 1960s had ear placement varying from 0.8 to 1.3 m and they were more resistant to root lodging than the standard Mexico-5 that was the best available variety at that time (Anon. 1963). In 1964, mass selections were made from five maize populations (Anon. 1966) and recurrent selection continued in some other populations to extract new varieties and parental inbred lines for the production of hybrid varieties. The brachytic gene was incorporated into some of the populations to improve resistance to stalk breakage and root lodging. Opaque-2 gene was also incorporated into the breeding populations to improve the protein balance of the grain by increasing the lysine content. In 1967, maize introductions from the maize project at Kitale, Kenya, were tested in Zaria. The result showed that the introduced materials were poorly adapted to the northern Guinea savanna conditions but performed much better at Gembu on the Mambilla Plateau, south of Yola, at an elevation of about 1566 m. Some of the Kenyan introductions also produced good yields at Mokwa. By 1967 maize production in northern Nigeria was becoming important in the riverine areas. For that reason, IAR Samaru posted a plant breeder to Mokwa to support the Federal Program on Maize (Anon. 1969). In 1969, IAR Samaru released SAMMAZ-6, formerly Biu Yellow. In 1972, three new varieties, namely SAMMAZ-7, SAMMAZ-8 and SAMMAZ-9 bred at IAR Samaru were also released. These varieties had potential yields ranging from 5 to 7 t/ha (Anon. 1989). By 1973, the research on maize at IAR Samaru resulted in the production of Extension Bulletin No. 11 titled "Maize Production Guide", which was modified later to "Recommended Practices for Maize Production." By 1976, IAR breeders were busy developing synthetic varieties at Samaru as well as Mokwa and Kadawa substations. Genetic studies were conducted with emphasis on yield and protein content.

In collaboration with the International Institute of Tropical Agriculture (IITA), the varietal improvement project continued in 1982 at IAR Samaru with the European Economic Community (EEC) Project coordinated by Dr Khadr. The EEC Project focused on the development of intermediate white maize populations (110 days to maturity) through recurrent selection, development of mid-altitude populations resistant to streak and other diseases (*Helminthosporium turcicum, Puccinia sorghi*) in Jos and development of disease resistant and vigorous inbred lines with good combining ability. The Project was successfully conducted in collaboration with the staff of IAR, Samaru (Anon. 1983).

Fakorede *et al.* (1993; 2001) had presented excellent reviews of the Nigeria-IITA hybrid maize project. Initiated in 1982, the Nigerian government funded the project, which was executed by IITA in collaboration with the Nigerian NARES, including IAR Samaru. The era of hybrid maize in Nigerian agriculture started in 1984 when hybrids were planted to a total of 150 ha of farmers' fields. The hybrids have contributed to increased productivity of maize, expansion of land area planted to maize and establishment of private seed companies in Nigeria. IAR Samaru has continued to play an important part in the development and release of hybrid maize in Nigeria. Several hybrid varieties were tested and released over the years (Menkir *et al.* 2001). Among the hybrids tested in the early stages of the project, the white hybrid 8321-18 gave the best yield and was, therefore, the first hybrid to be released in 1985. Many other hybrids have since been developed, although the hybrids were susceptible to *Striga hermonthica* Del Benth, a parasitic weed of maize and several other crops in the savanna ecology.

Variety testing was initiated as a sub-project at IAR in 1966. Seeds of several mid-to-high altitude varieties from Kenya were planted on the Mambilla Plateau. The varieties showed greater potential for the area than the local varieties. IAR participated in the Regional Uniform Variety Trials (RUVTs) of the Semi-Arid Food Grains Research and Development (SAFGRAD) Project conducted in the late 1970s to early 1980s, tailored to develop suitable varieties for areas with shorter and unpredictable rainfall duration. Similarly when WECAMAN was established in 1987 to document maize production constraints and available resources in the region and to conceive a strategy to pool their resources with those of IARCs for the development of appropriate maize production technologies (Fajemesin *et al.* 1999), IAR was fully involved as an active collaborator. In 1984, the Nationally Coordinated Maize Research Programme (NCMRP) was initiated to link together research, production, industry and consumers (Fakorede 1993). NCMRP organizes the testing and release of superior varieties of maize with higher yield potential and better resistance to insect pests and diseases. Since 1990, IAR Samaru has been mandated to coordinate maize research in Nigeria.

Seed Production

Seed production is crucial to the transfer of improved maize technologies to farmers; therefore, IAR initiated a seed production sub-project in the 1960s. In the early 1960s, the multiplication and distribution of improved maize varieties from IAR took place substantially in five provinces as follows: ESI in Niger, Adamawa, Benue and Plateau Provinces; Mexico-5 in Plateau; ESII and EAAFRO 231 in Ilorin. In 1962, ESI was multiplied at Osara and Ochanja in Kabba Province as an early maturing variety and distributed to selected farmers who grew it as a second season crop. Other areas where maize seed multiplication was carried out were Zaria, Bornu and Katsina Provinces.

IAR has been a collaborator in the WECAMAN sponsored community-based seed production project since 1993 and large quantities of seed of released varieties are produced annually in Nigeria, among other WCA countries such as Bénin, Burkina Faso, Cameroon, Mali and Togo (Badu-Apraku *et al.* 1999). In 2001, in collaboration with IITA scientists, SAMMAZ 11, a *Striga hermonthica* resistant open-pollinated variety, was registered and released in Nigeria along with two extra-early varieties, SAMMAZ 12 and SAMMAZ 13 (Ado *et al.* 2002a,b).

Agronomic Research

The contributions of IAR in the area of cultural practices and management of maize are numerous and well documented. Several published and unpublished research reports on maize agronomy are available at IAR Samaru. Agronomy research in 1959 indicated that local maize varieties had low yield potential. Trials on inter-planting maize with sorghum in the riverine areas were conducted. Fertilizers were applied to maize, which was later inter-planted with sorghum that served as a residual crop. By harvesting the two crops in one year, marginal economic responses were obtained. Results of the 1959 fertilizer trials indicated a positive linear interaction between sulphate of ammonia and single super phosphate. By 1963, sufficient information had been obtained from the annual fertilizer trials that led to the conclusion that the northern part of the maize growing area of the savanna had a relatively high response to P while the southern part, which is the derived savanna, had a relatively high response to N. By 1965, accurate estimates of the requirements of N and P for sorghum and maize in the main growing areas of northern Nigeria were provided to the Ministry of Agriculture for recommendation to farmers. Recommendations had also been made to the Ministry of Agriculture on the use of compound fertilizer to replace separate application of single super phosphate and sulphate of ammonia. This would have the advantage of increased efficiency in the use of plant nutrients and of considerable savings in transportation and handling costs. Also uniform application of fertilizers at moderately high rates was found necessary to decrease variability in plant development and to measure the yield potential of the most productive varieties or hybrids.

At Mokwa, research results showed that K application up to 90 kg K2O/ha did not increase grain yield significantly. In some other trials no significant response to K2O and Zn was found in the savanna zones. However, application of K2O increased grain yield at Yandev (southern Guinea savanna, SGS), Ballah (SGS) and Kafin Maiyaki (northern Guinea savanna, NGS). In another fertilizer trial, 30 to 60 kg K2O/ha appeared optimum in the SGS while 2.5 kg Zn/ha gave maximum grain yields at Ballah and Kafin Maiyaki (Anon 1987). In a trial conducted for two years, there was no response to Zn at Yandev. In 1984, a linear response to N from 0 to 120 or 150 kg N/ha was reported in some studies, although the application of more than 90 kg N/ha did not result in significant increases in grain yield in much of the savanna ecology. Ear length and some other traits also responded linearly to N fertilizer up to 120 kg/ha. Split N applications did not, however, appear to enhance maize grain yield. The requirements for K and Zn increased with increased rainfall and clay content of the soil.

Data on planting date trials at Samaru revealed significant yield reduction for delayed plantings after early June. Plant spacing trials gave similar yields if plantings were one plant spaced 25 cm apart in the rows, two plants on hills spaced 50 cm apart or three plants on hills spaced 75 cm apart. In another trial conducted at Kadawa, three varieties SZV5, SZV3 and DMR-ESR-Y were compared with Bomo local in four planting dates. The interaction between sowing dates and varieties was not significant. The highest yield was, however, obtained when sown early in July. Decrease in grain yield associated with delayed planting was minimized when the plant density was reduced. The delayed planting resulted in lower yield as well as lower attack by ear rot organisms. Increasing plant density up to 100,000 plants/ha at early planting increased grain yield and other yield parameters. In another trial, a 26% yield increase was obtained when plant density was increased from 25,000 to 50,000 plants/ha. In this particular study, increasing plant density beyond 50,000 plants/ha had no appreciable effect on grain yield.

Early planting provided an escape to *Striga* attack because the *Striga* plants were just emerging when the maize had already established. Out of six hybrids tested, hybrid 8322-13 was the most resistant to *Striga*. In 1990, Weber *et al.* (1995) reported yield and *Striga* infestation on susceptible and *Striga* tolerant versions of maize varieties in 13 fields in Kaduna and Katsina States. Herbicides experiments showed that at Samaru, acetochlor at 0.5 kg a.i/ha, plus atrazine at 1.25 kg a.i/ha was the best treatment for weed control. Primagram gave the highest grain yield in comparison with other pre-emergence herbicides and herbicide mixtures.

Crop Protection

Research on crop protection revealed that in 1964 insect infestation associated with virus diseases prevented maize seeded in July from flowering at the Kagoro Ranch near Kafanchan. In the same year an epidemic of maize rust caused by *Puccinia polysora* (Anon 1966) developed in August and September in the Samaru experimental nurseries. A detailed study of virus diseases of maize and other grass host plants was subsequently advocated in Northern Nigeria. On stem borer damage, it was reported that maize stubble was not a source of carry-over of the two main stem borers *Sesamia calamistis* and *Eldana saccarina,* to the subsequent cropping season (Anon. 1969). Disease surveys conducted in 1983 indicated that maize streak virus (MSV), Drechslera (*Helminthosporium maydis*) and the yellow blotch bacterium were the major causal agents of diseases in maize. Rust and leaf spot diseases were severe in higher rainfall areas. Infestations by *Striga hermonthica* and stem borers were other biotic problems. Early sowing was of advantage against stem borers. However, sowing date did not significantly reduce infestation by *S. calamistis* and *E. saccarina*.

MSV affected 30% of the crops around Samaru but in areas north of Samaru the incidence was up to 60% probably because of late planting, which was due to late rains in 1984. Apart from host plant resistance, soil treatment with Furadan 3G was the most effective against the MSV disease under field conditions. Stem borers were of greater economic importance than earworms. By 1985, it was reported that granular carbofuran applied into the furrow at planting at the rate of 1.0 kg a.i./ha gave the best protection against the MSV disease and resulted in the highest grain yield. In 1986, the reaction of 38 exotic and local varieties of maize to four major diseases *Drechsler* leaf blight, MSV, rust and *Curvularia* (Anon 1987) were highly variable, suggesting the possible occurrence of different pathogenic strains. Granular and seed treated formulation of Furadan effectively controlled MSV disease and produced the highest grain yield. In another trial, 3% granular formulation of the systemic insecticide, carbofuron, applied at the rate of 1.0 kg a.i/ha to the furrow at planting gave the best protection for maize, the highest grain yield and the highest monetary return. On grain storage, loss in weight was significantly lower with Actellic EC and with Sumithion and Reldan at dosage rates of 5 and 10 ppm, respectively. Ear rot disease damage was least where maize was left to dry in the field. On nematodes control, both Furadan and Counter resulted in significant increase of maize grain yields. Sumithion treated maize grain was found to have the least numbers of *Sitophilus tribolium* and *Oryzaephilus* spp. in an on-farm trial.

Economics of Production

On economics of production and technology adoption, research at IAR revealed that new maize technology packages gave 98.8% higher grain yield than the farmers' traditional methods of production. All the components of the new technology contributed to increased grain yield. The new technology was, therefore, cost-effective. Although the labour requirements for the two types of maize (OPV and hybrids) were similar, results of on-farm trials indicated that, on average, hybrids out-yielded an OPV, TZB, by 53%. Therefore, there is a significant financial advantage of growing hybrids over TZB. At high rates of fertilizer application, hybrid varieties gave higher gross margin; however, at lower rates of fertilizer, TZB gave a higher gross margin in 1989. Under the lowest fertilizer rates, however, average net revenue from hybrid production was, on average, about 36% higher than that of the OPV.

Collaborative Trials

IAR has been collaborating with national and international research organizations on maize improvement since the 1960s. By 1962, IAR was fully collaborating with the Federal Department of Agriculture on maize research. In 1965, the IAR maize project was expanded to include not only the National Co-operative Trials, but also additional yield trials of varieties and hybrids. Mexico-5 used as a standard check at Samaru was only exceeded in yield by H5O3. By 1967, IAR researchers were conducting the National Zonal Trials as well as the West African Yield Trials in collaboration with OAU-STRC Joint Project with the seed for the trials provided by six countries. In 1975, IAR Samaru co-operated in the East African Maize Variety Trial Co-ordinated by East African Agricultural and Forestry Research Organization (EAAFRO), Kitale, Kenya. Regional Uniform Variety Trials (RUVTs) were used as the vehicle for testing the performance of elite varieties under different environmental and socioeconomic conditions and for the direct exchange of varieties and source germplasm among countries. The goal of the collaborative trials was to afford national scientists the opportunity to identify varieties for broad or narrow adaptation, as appropriate for their individual situations.

In 1983, in collaboration with IITA Ibadan, inbred lines resistant to MSV and selected hybrids developed from them were tested in comparison to the commonly grown OPVs. The hybrids were superior to the OPV checks. In 1988, seven entries of the Nationally Co-ordinated Trials (namely TZE Comp 3, Funtua 86 TZU TSR-W, Across 86 TZUTS R-7, EV8444-SR, Maracay 7921-SR, EV8422-SR and Farako Ba 85TZSR-Y-1) performed very well. For the Regional Trials, one entry (Pop CSP early) out of the 12 extra-early maturing varieties showed outstanding performance. Among the 12 medium maturing entries, 8 significantly out-yielded the check, with EV8444 SR producing the highest yield. In collaboration with NCRI and IITA, IAR contributed significantly to the development of improved varieties for different ecological zones. Some of the varieties released include TZA, TZB, TZESR-W, TZESR-Y and Bulk3. In 1989, the Nationally Coordinated Trials continued with 15 elite cultivars with TZESR as check. Also 13 SAFGRAD varieties included in a Regional Trial were planted at two locations in the NGS and one location in the SS. Among the early maturing varieties, Across 86 Pool 16 DT, significantly out-yielded the other 13 entries. Among the extra early materials, entries TZESR-W × Gua 313 BCF6 and TZEF-Y performed significantly better than most of the other entries.

The SG-2000 started activities in Nigeria in March 1992 to raise agricultural productivity and improve food marketing. SG-2000 initially concentrated on maize and wheat in Kano and Kaduna States but now has moved to other crops and other States of the Federation (Valencia and Breth, unpublished). In recent years high-yielding, early and extra-early maturing disease resistant maize, along with fertilizer became available, leading to the cultivation of the crop in "non-traditional" areas, such as some parts of the Sudan and Sahel savannas. The collaboration of SG-2000 with IAR Samaru, NARIs, IITA and States ADPs for technology transfer approach resulted in doubled or tripled crop yields over the national average. This has led to a marked improvement in income and living standards of farmers in the savanna zones of the country. SG-2000 in partnership with IAR, IITA, WECAMAN and CIMMYT is now promoting quality protein maize (QPM) in much of the savanna zone of Nigeria.

Impact of Maize Research

Van Eijnatten (1965) predicted that with improved transport facilities, the farmers in northern Nigeria would intensify the production of maize for export to southern areas. Nigeria does not need to import any type of maize whatsoever; on the contrary it could be a maize exporting country. Land area planted to maize and yields obtained per ha from 1982–2002 are presented in Fig. 1. The land areas under maize increased at about 2,800 ha per year from 1982 to 2000.

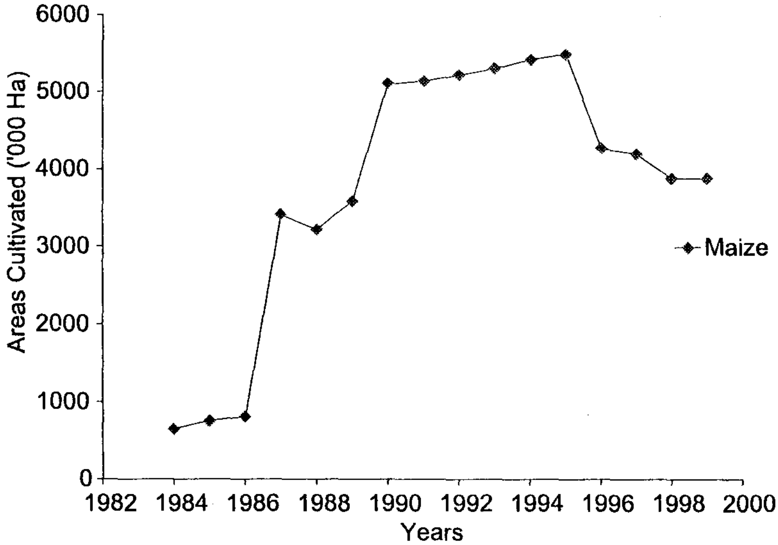

Figure 1. Trend of land area cultivated to maize in Nigeria, 1984–2000.

There has been a dramatic increase in the land area under maize in the northern Guinea savanna (Fakorede *et al.* 1993), which may now be described as the Corn Belt of Nigeria. Availability of fertilizers at affordable prices generally determines the increase in land area under maize production in any particular year. Thus, areas cultivated to maize decrease as fertilizer subsidies are withdrawn. When fertilizers were not readily available, the land put to maize was reduced because maize production depends on availability of fertilizers.

The trends for grain yield and production were similar to those of land area (Fig. 2). Average annual increase in total production was much higher than the annual increase in yield.

The average maize grain yield increased from 0.9 t/ha in 1980 to about 1.3 t/ha in 2001, an annual increase of 0.02 t/ha. Between 1980 and 2001, land area under maize increased significantly up to 440% (Fakorede *et al.* 2003).

Maize is rapidly being adopted in many marginal areas because of availability of early and extra-early maturing varieties. Grain yield has shown significant linear increases as a result of adoption of the improved technologies. Collaborative efforts in RUVTs, on-farm demonstrations, capacity building, exchange of ideas and technical experience of NARIs and international scientists, and promotion of community-based seed production have all contributed to the increased maize output (Fakorede *et al.* 2003). According to these researchers, estimated annual growth rate in maize production was 2.76%, which is lower than the 3.20% projected to meet local demands (Shaib *et al.* 1997). Despite the increase in production, the demand for maize is higher than even the target set for the attainment of selfsufficiency in the country (Fig. 3). In order for Nigeria to be selfsufficient in maize

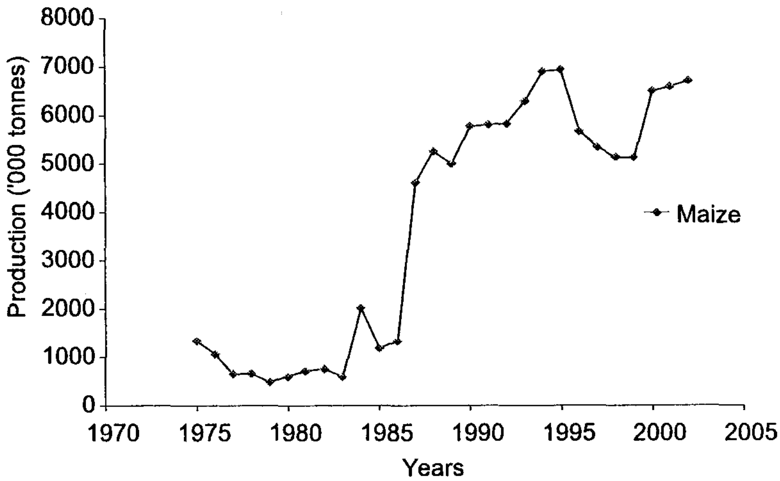

Figure 2. Production trend for maize in Nigeria, 1975–2005.

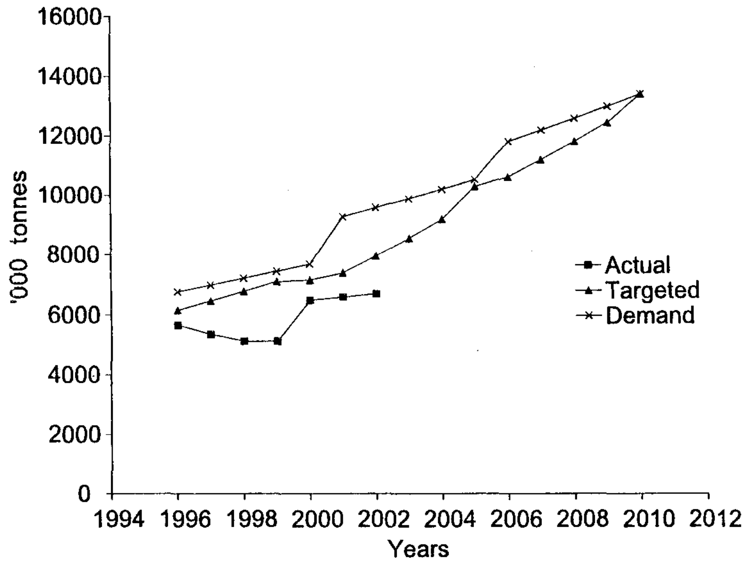

Figure 3. Trends of actual production, targeted production and projected demand for maize in Nigeria.

production, it must produce enough for consumption and have surplus for export. To satisfy this demand, the nation must be able to produce a minimum output of 10 million t annually (Azeez 1993).

Social gains from maize research in the northern Guinea savanna (NGS) of Nigeria during the 1986–97 period indicated varietal adoption rates of 7.8%, 49.6% and 42.6% per annum for local, improved OP and hybrid varieties, respectively, with an internal rate of return of 23%. Thus, the investments in maize research were well justified (Phillip 2001). Poor yields in farmers' fields were caused by low soil nutrients and poor management practices (Fakorede *et al.* 2001). To realize maize yield potential, there should be a combination of appropriate variety, soil and crop management practices.

Conclusion

In conclusion, the Institute for Agricultural Research, Samaru, Nigeria, together with international and other national research institutes and universities, is fully committed to the development of improved high yielding varieties, appropriate technologies and sustainable production systems. Through the collaboration of these institutions, land area under maize, total maize production and grain yield per ha have increased dramatically in the last two decades. In achieving this level of success, the particularly fruitful collaboration of IAR with IITA, WECAMAN, CIMMYT, SG-2000 and the Nigerian NARES deserves special mention.

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